CIRCULAR MOTION — DPP 01

Single Correct Answer · Multiple Correct Answers

SECTION A — Single Correct Answer Type

Only one option is correct.

1. A wheel having a diameter of 3 m starts from rest and accelerates uniformly to an angular velocity of 210 rpm in 5 s. Angular acceleration of the wheel is

(A) 1.4 π rad s^{-2} (B) 3.3 π rad s^{-2}
(C) 2.2 π rad s^{-2} (D) 1.1 π rad s^{-2}

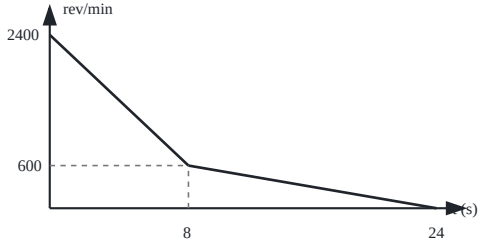
2. A body is thrown with the velocity v_0 = 10 m s^{-1} at an angle of $\alpha = 37°$ from the horizontal ground. Find the angular velocity of the body as observed from the point of projection at the time of landing.

(A) 5/8 rad s^{-1} (B) 5/6 rad s^{-1}
(C) 5/3 rad s^{-1} (D) None of these

3. A particle moves in the xy-plane. The position vector of the particle at any time t is r = (2t i + $2t^2$ j) m. θ is the angle which its velocity vector makes with the positive x-axis. The rate of change of θ at time t = 1 s is

(A) 3/7 rad s^{-1} (B) 2/5 rad s^{-1}
(C) 2/17 rad s^{-1} (D) None of these

4. A table fan, rotating at a speed of 2400 rev/min, is switched off and the resulting variation of the revolutions per minute with time is shown in the figure. The total number of revolutions of the fan before it comes to rest is



(A) 280 (B) 140 (C) 70 (D) 210

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Single Correct Answer · Multiple Correct Answers

SECTION A — Single Correct Answer Type (Continued)

Only one option is correct.

5. In uniform circular motion

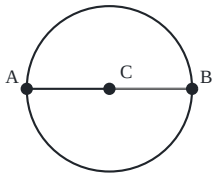
(A) both velocity and acceleration are constant
(B) acceleration and speed are constant but velocity changes
(C) both acceleration and velocity changes
(D) both acceleration and speed are constant

6. A particle is revolving in a circle increasing its speed uniformly. Which of the following is constant?

(A) Centripetal acceleration (B) Tangential acceleration
(C) Angular acceleration (D) None of these

7. A particle P is moving in a circle of radius ‘a’ with a uniform speed v. C is the centre of the circle and AB is a diameter. When passing through B the angular velocity of P about A and C are in the ratio

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1



10. When a ceiling fan is switched off its angular velocity reduces to 50% while it makes 36 rotations. How many more rotations will it make before coming to rest (Assume uniform angular retardation)?

(A) 18 (B) 12 (C) 36 (D) 48

SECTION B — Multiple Correct Answers Type

One or more options may be correct.

11. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle. The motion of the particle takes place in a plane. It follows that

(A) velocity is constant
(B) acceleration is constant
(C) kinetic energy is constant
(D) it moves in a circular path